

MPG No. 27.8.156 $6'/mmm'$ ($6'/mm'm$ setting) [Type III, hexagonal] [M tensor]

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss). * Rank 3 tensor (sa). * Rank 3 tensor (a). * Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xyz} & 0 & 0 \\ 0 & 0 & 0 & -M_{xyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ M_{xyz} & -M_{xyz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{xyz} \\ 0 & 0 & 0 & 0 & M_{xyz} & 0 \end{bmatrix}$$

$$M_{xyz} = \mathbb{M}_{4a}^{(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xyz} & 0 & 0 \\ 0 & 0 & 0 & -M_{xyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -M_{xyz} & M_{xyz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -M_{xyz} \\ 0 & 0 & 0 & 0 & M_{xyz} & 0 \end{bmatrix}$$

$$M_{xyz} = 2\mathbb{T}_{3a}^{(1)}$$

* Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} M_{xyz} & 0 & 0 \\ -M_{xyz} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & -M_{xyz} & 0 \end{bmatrix}$$

$$M_{xyz} = \mathbb{T}_{3a}^{(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} M_{yzxx} & -M_{yzxx} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -M_{yzxx} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzxx} = \mathbb{T}_{3a}^{(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xyz} & 0 & 0 & M_{xxzy} & 0 & 0 \\ 0 & 0 & 0 & -M_{xyz} & 0 & 0 & -M_{xxzy} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ M_{yzxx} & -M_{yzxx} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{yzxx} & 0 & 0 & M_{yzxx} \\ 0 & 0 & 0 & 0 & M_{xxzy} & 0 & 0 & M_{xyz} & 0 \end{bmatrix}$$

$$M_{xyz} = \mathbb{M}_{4a}^{(1)} + 2\mathbb{T}_{3a}^{(1)} + \mathbb{T}_{3a}^{(2)}$$

$$M_{xxzy} = \mathbb{M}_{4a}^{(1)} + 2\mathbb{T}_{3a}^{(1)} - \mathbb{T}_{3a}^{(2)}$$

$$M_{yzxx} = \mathbb{M}_{4a}^{(1)} - 2\mathbb{T}_{3a}^{(1)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} M_{yzxx} & -M_{yzxx} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -M_{yzxx} & 0 & 0 & -M_{yzxx} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzxx} = \mathbb{T}_{3a}^{(3)}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & M_{yyyyz} \\ 0 & 0 & 0 \\ 0 & M_{yyyyz} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -M_{yyyyz} \\ 0 & 0 & 0 \\ 0 & -M_{yyyyz} & 0 \\ 0 & 0 & 0 \\ -M_{yyyyz} & 0 & 0 \end{bmatrix}$$

$$M_{yyyyz} = -\mathbb{M}_{4a}^{(1)}$$